MIL-S-43032A(GL)
28 February 1967
SUPERSEDING
MIL-S-43032(QMC)
30 June 1961

MILITARY SPECIFICATION

SAW, HAND, CROSSCUT, 24-INCH BLADE

1. SCOPE

* 1.1 This specification covers one type and size crosscut hand saw having a 24-inch blade.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-H-15424 - Hand Tools; Packaging of

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

DRAWING

* ARMY NATICK LABORATORIES

4-5-13 - Saw, Hand, Crosscut, 24-inch Blade

(Miniature copy of Drawing 4-5-13, identified as figure 1, is attached for information purposes only.)

FSC 5110

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* (Copies of specification, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES DEPARTMENT OF COMMERCE

Commercial Standard GS35-61-Hardwood Plywood

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C. 20402.)

3. REQUIREMENTS

3.1 Preproduction sample.- When specified (see 6.2), before production is commenced, a sample of the finished saw shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of the responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for manufacture of the production items.

3.2 Materials.- Materials specified shall conform to the applicable specifications, standards and requirements specified herein. Materials not definitely specified shall be of the quality normally used for the purpose in commercial practice.

* 3.2.1 Steel.- The steel for the saw blade shall be hot-rolled saw steel and shall have a hardness of Rockwell C48 to C52 when tested as specified in 4.4.1. The steel shall be free from cracks, segregations, inclusions, seams, hammer marks, high spots, or other imperfections.

* 3.2.2 Plywood, hardwood.- Hardwood plywood shall be 7-ply minimum, consisting of all hardwood veneers of high density, in accordance with type I, grade 1 of Commercial Standard CS35-61.

3.3 Construction.- The saw shall be constructed in accordance with the details and dimensions shown on Drawing 4-5-13, and as specified herein.

3.4 Component details.-

3.4.1 Blade.- The blade shall be made from steel as specified in 3.2.1. When tested as specified in 4.4.2, there shall be no permanent set in the saw blade.

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3.4.1.1 Teeth.- The cutter teeth shall be filed to a cutting point and beveled to an angle of approximately 30 to 40 degrees with the face of the saw. The cutter teeth shall be set not more than one half of the depth of the teeth and as shown on Drawing 4-5-13. Raker prongs shall be filed to a cutting edge square with the center plane of the thickness of the saw. The raker teeth shall not be set or swaged.

* 3.4.2 Handle.- The handle shall be made from hardwood plywood specified in 3.2.2. The handle shall be secured to the saw blade by not less than four attaching devices specified in 3.4.3 in such a manner that there shall be no looseness between the handle and blade. Holes provided for the attaching devices shall be countersunk or counterbored in such a manner that the devices will be flush with the surface of the handle.

3.4.3 Attaching device.- Attaching devices shall be made of metal. Attaching devices shall be capable of detachment for replacement of handle. Ferrous metal attaching devices shall be either zinc, cadmium, or nickel plated.

3.5 Finish.-

3.5.1 Blade.- The surface of the saw blade shall be polished in accordance with the manufacturer's standard practice. There shall be no thinning or "rubbing off" near the edges of cutting teeth or rakers.

3.5.2 Handle.- Wood handles shall have polished and rounded edges. All exposed surfaces, including the inside of the hand grip, shall be finished smooth, stained, and either wax coated or coated with a clear lacquer or varnish. The finish shall be transparent to the extent of not concealing the grain of the wood.

* 3.6 Marking for identification.- The blade of the saw shall be permanently marked with the letters U.S. in characters not less than 1 inch high and the manufacturer's name or trademark of such known characters that the source of manufacture may be readily determined.

3.7 Workmanship.- The finished saw shall be clean, well-made, and free from rust and any other defects which may affect appearance or serviceability. Other than the cutting surface, the blade and handle shall be free from burrs and any projections which might cause injury to the hands of the operator.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial

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laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection.- When a preproduction sample is required, it shall be examined for defects in table I, dimensions specified, and tested as specified in 4.4.2.

4.3 Sampling for inspection.- Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Intermediate test.- The blade shall be tested for hardness in accordance with 4.4.1. Three determinations shall be made on each sample unit. Failure of one or more determinations to meet specified requirements shall be classified a defect. The inspection level shall be S-3 with an acceptable quality level (AQL) of 2.5 defects, expressed in terms of defects per hundred units. The lot shall consist of all blades offered for inspection at one time. The sample unit shall be one blade.

4.3.3 End product inspection.- The lot shall consist of all fabricated saws offered for inspection at one time. The sample unit for the inspection shall be one completely fabricated saw.

4.3.3.1 Visual examination.- Examination shall be made for defects listed in table I. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish: Blade	Not polished	X	
	Evidence of rust		X
Finish (coating): Attaching devices	Missing	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Finish: Plywood handle	Flaking, peeling or blistered	X	
	Not stained and coated with wax, lacquer or varnish	X	
	Finish not transparent (conceals grain of wood)		X
	Not smooth		X
	Finish not dry (transfers to hands)	X	
	Area not finished		X
	Dirt, grit or foreign matter imbedded in finish		X
Construction and workmanship general (applicable to all components and assemblies)	Handle not securely attached to blade as specified		X
	Handle split, cracked, or delaminated between plies, not required number of plies or sharp or rough edges	X	
	Burr or sliver on handle	X	
	Component missing (except attaching devices)	X	
	One attaching device missing		X
	More than one attaching device missing	X	
	Component fractured, cracked, bent, or malformed	X	
	Blade shows evidence of seams	X	
	Blade shows evidence of thinning or rubbing off near teeth edges, hammer marks, high spots, segregations, dents, or inclusions		X
	Tooth broken, teeth not filed as specified, chipped or cracked	X	
	Cutter teeth set more than one-half the depth of the teeth	X	
	Raker prongs or cutter teeth not filed as specified	X	
	Raker teeth set or swaged	X	
	Back edge of saw blade contains burr or sharp edge which may cause injury	X	
Marking	Missing, illegible, incomplete, incorrect, not permanent, or not applied as required		X

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4.3.3.2 Dimensional examination.- Inspection shall be made of the end item for compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects, expressed in terms of defects per hundred units.

4.3.3.3 Testing of the end item.- The end item shall be tested as specified in 4.4.2. The inspection level shall be S-3. Any failure shall be cause for rejection of the lot.

4.3.4 Examination of preparation for delivery.- Examination shall be made to determine compliance with preservation, packaging, packing and marking requirements in accordance with MIL-H-15424.

4.4 Tests.-

4.4.1 Hardness.- Tests shall be performed in accordance with method 243.1 of Fed. Test Method Std. No. 151 for compliance with 3.2.1.

4.4.2 Bending test.- The saw blade shall be bent slowly around a wooden mandrel 12 inches in diameter until the ends of the blade form a 90 degree angle. The bend shall be made at the midpoint of the length of the blade. After the above procedure, the saw blade shall be reversed and bent in the opposite direction. The saw blade shall be bent alternately through three cycles, after which the saw blade shall be examined to determine compliance with the requirements specified in 3.4.1.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C.- Saw shall be preserved, packaged, packed and marked for each level of protection in accordance with the applicable requirements of MIL-H-15424 (see 6.2).

6. NOTES

6.1 Intended use.- The saw covered by this specification is intended to be used by small tactical units to cut down trees up to 12 inches in diameter, also limbing to blaze trails, including frozen trees; for cutting holes in ice, and for cutting blocks of compacted snow.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) When preproduction sample is required (see 3.1)
(c) Selection of applicable levels of preservation, packaging, and packing (see 5.1)

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6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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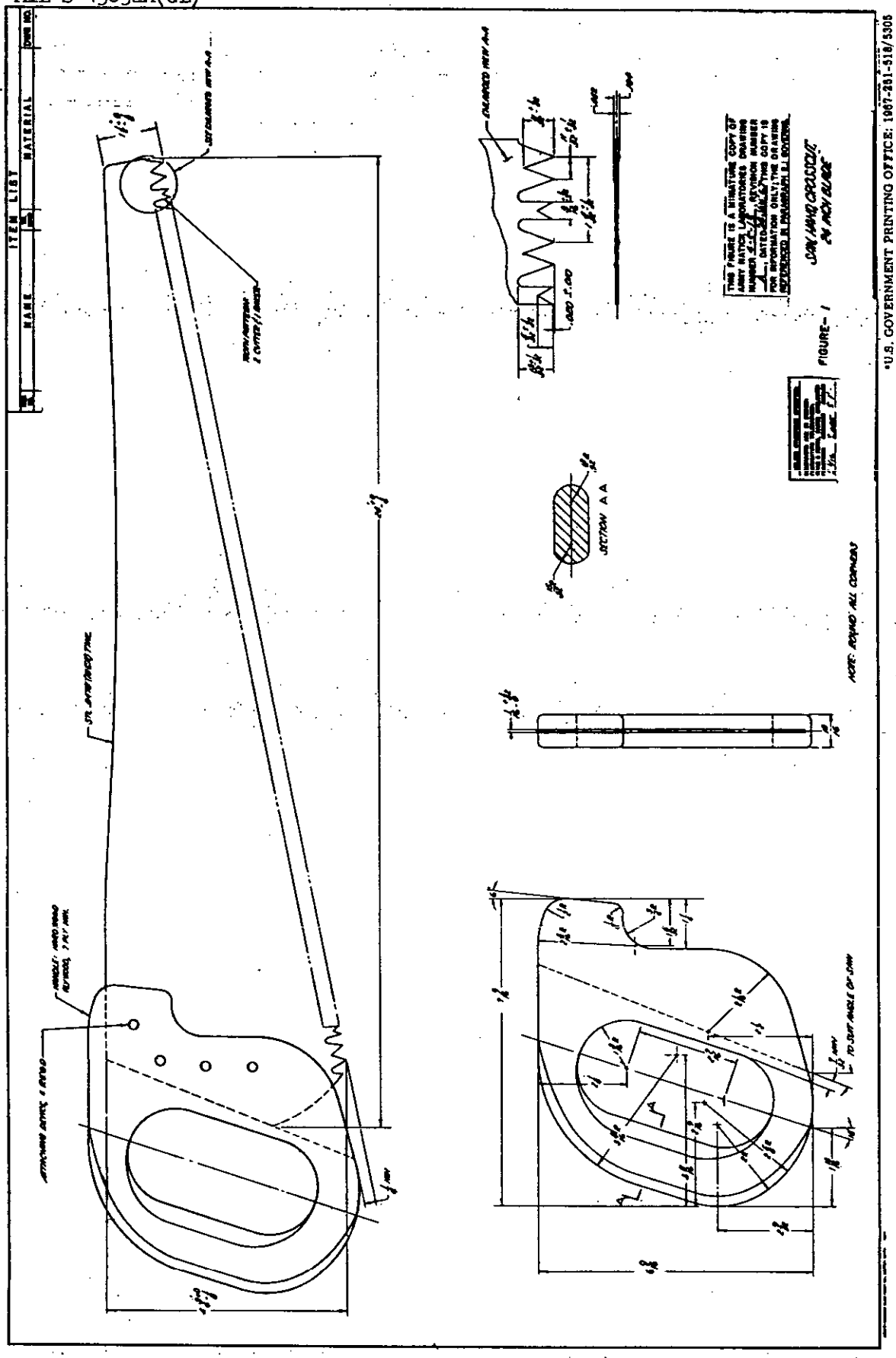
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Saw, Hand, Crosscut, 24-Inch Blade MIL-S-43032A(GL)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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